

Sentinel

Every answer comes with its evidence.

A document checker that is confidently wrong is worse than no checker at all - because nobody goes back and looks.

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duocode-sentinel.vercel.app · github.com/TCF1209/duocode-sentinel

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The problem - and the fourth case

01 Classify

Tell message types apart: comparison requests, new SI requests, invoice queries, general mail, spam.

02 Extract

For comparison requests, read the SI and BL attachments and pull the matching shipment fields.

03 Compare

Surface any mismatched fields, showing the SI and BL values side by side.

04 Ask for help

When it can't finish on its own, escalate to a person with full context instead of guessing or failing silently.

Why we built it: today a documentation clerk does this manually - about 4 minutes per SI/BL pair and 20 seconds to triage each email (a conservative estimate, not a measurement) - and a missed field becomes a correction, a delay, rework.

Finding the right emails takes time

A document request that is overlooked never reaches the checking step.

Manual comparison is easy to get wrong

Names, ports, quantities and weight across two documents; a missed discrepancy is a correction, a delay, rework.

The same fact looks different

One document says 'Port of Loading', the other 'Load Port' - the system has to know they are the same field.

The fourth case: sometimes the check cannot be done at all - an unreadable scan, a blank field, the wrong document. That has to be escalated to a reviewer with the reason attached, not guessed at. Sentinel does all four.

What it does, end to end

Sentinel

HomeCompareRunsHow it works

Go to Runs →

← Back to Runs

run_000002_1790314413

Done

520/520 emails processed

Before Sentinel

With Sentinel

Patterns

View metrics

Sorted into

All 520

Comparisons 220

SI requests 125

Invoice queries 75

General 60

Spam 40

Outcome

All 520

No discrepancy 454

Discrepancy 46

Escalated 20

Reviewed 0 / 220 comparisons

100% by rules

6 model calls

Review

All

Order

Needs attention first

520 cases

⚠ Patterns worth a second look (6)

Show >

EMAIL	CATEGORY	CONFIDENCE	OUTCOME	DISCREPANT FIELDS	DECIDED BY
email_004	Comparison	100%	Discrepancy · 2 fields	Consignee, Notify Party	RulesOpen
email_025	Comparison	100%	Discrepancy · 2 fields	Container Count, Port of Discharge	RulesOpen
email_031	Comparison	100%	Discrepancy · 2 fields	Container Count, Gross Weight (kg)	RulesOpen
email_065	Comparison	100%	Discrepancy · 2 fields	Notify Party, Port of Discharge	RulesOpen
email_071	Comparison	100%	Discrepancy · 2 fields	Container Count, Port of Discharge	RulesOpen
email_097	Comparison	100%	Discrepancy · 2 fields	Container Count, Gross Weight (kg)	RulesOpen
email_107	Comparison	100%	Discrepancy · 2 fields	Consignee, Container Count	RulesOpen

520

emails, classified and routed

46

discrepancies, with the fields that differ

20

escalated - with the reason and what to do

12.7 s

for the whole inbox, on a free-tier container

How a desk uses it: 1 Run the inbox · 2 Open a flagged case · 3 Confirm or override it - correct a single value or attach the amended SI/BL · 4 Send the drafted reply

How it decides



Six stages, one direction. The gate runs after the comparison: a value we cannot find again in the document it was read from is never reported as a discrepancy - it is escalated to a reviewer, with both readings attached.

Labels by meaning, values exactly

Never a similarity score. A threshold loose enough to forgive a scan artefact also merges two real companies - and this data has them.

Rules first, model second

Every stage tries a deterministic answer before a model, and records which one answered (decided_by). The model is asked only where the rules admit they cannot read.

One stateless pipeline library

No web framework or database in backend/sdoc. The CLI, the API and the tests run the same code, so the thing scored is the thing demonstrated.

Scaling, as built: state sits behind one class in one file (store.py) - Postgres is a one-file change · one worker on purpose · throughput is more copies of a stateless library · cost does not scale with volume on this inbox · per-desk rules slot into the existing stage boundaries.

Where the AI is - and why it is aimed



Classify - an email the rules cannot separate

asked only when the rule score is ambiguous; a closed five-way answer, never free text



Read - a field label the table has never seen

asked only about fields no rule resolved; every value must be found again in the document before it is accepted



See - a scanned page with no text layer

transcribed for the reviewer, never fed into a comparison - the case stays escalated

168 → 2

cases escalated on wording we invented - rules alone vs rules + model, 188 documents

False discrepancies: 0 both ways. Recall bought by guessing would have shown up there; it didn't.

gpt-5-mini, structured output. One rule for all three: nothing the model returns is accepted until it is found again in the document.

\$0.0013 per document at the published rates · cached · \$2 ceiling per run. On this inbox: 0 classifier calls, 0 extractor calls, 6 scans transcribed for the reviewer - every decision is a rule's, and that is measured, not assumed.

How we know it holds

Not memorised

1.0000 on four datasets, three from seeds we never developed against - 225 planted discrepancies caught with the exact field set, 80/80 escalations correct.

Attacked ourselves

16 kinds of damage, 3,008 perturbed documents, 20,496 field reads, no answer key. Thirteen modes at zero movement; silent wrong values 982 -> 0.

A real carrier's form

CMA CGM's public SI template, from outside the generator: four fields held, one bug found and fixed the same day.

Engineering

733 tests, 0 failing; CI on every push; a container that runs as a non-root user with no secret baked in; every value carries its evidence.

What we haven't fixed

email_145: a wrapped party name cut short to exactly what the other document says leaves the repair nothing to repair, and one real discrepancy in 3,008 perturbed documents is reported as consistent.

The obvious guard would flag 114 of 124 SI/BL pairs (92%) - worse than the gap - so it stays open and written down (ADVERSARIAL.md 5.2).

A defect we hide is worse than one we miss.

A perfect score on the dataset you were handed proves you didn't memorise it. It doesn't prove the reader works - so we went looking for the failures ourselves.

Live demo

90 seconds, on the deployed URLs

1

Start a run with the model tier on - then Before / With Sentinel: the inbox as it arrived, and what it made of it

2

Patterns worth a second look - one shipper, one field, seven times

3

A discrepancy case - under every value, the line it was read from

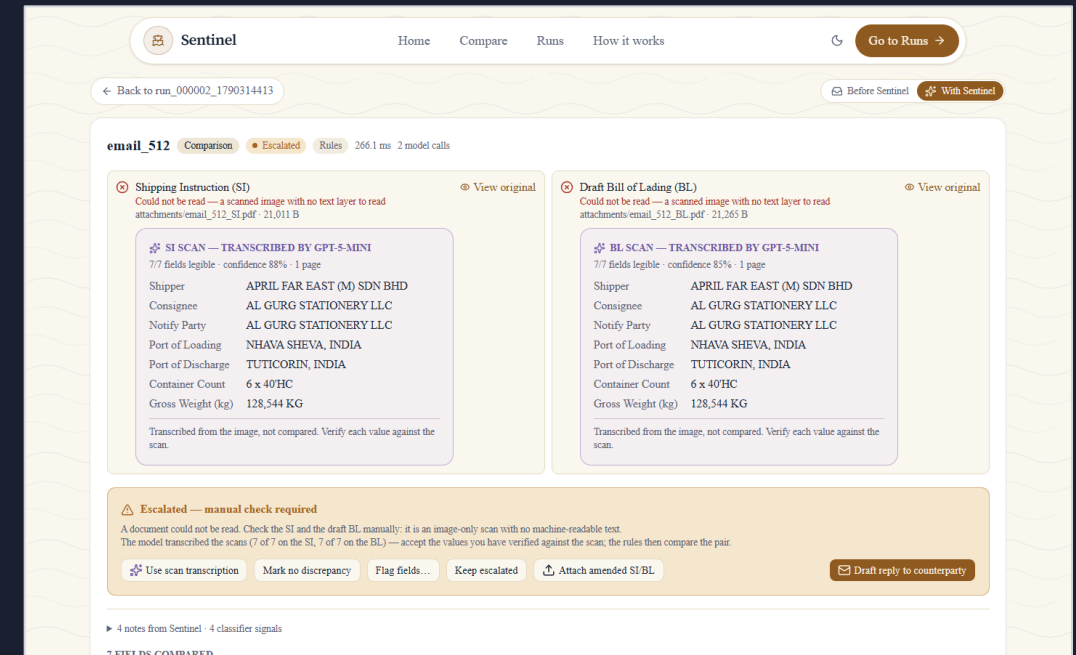
4

A scanned case - the page transcribed for the reviewer; the case still escalated, decided per field

5

Compare, unfamiliar labels: model off, then on - 'model answered'

Fallback if the venue network fails: the same inbox on the laptop, 1.5 s, no network - only the model beat changes.



What makes it different

- 1

Re-check on amendment

The sender amends the SI or BL? Attach the amended document on the case and the same check runs again. The previous result stays on record.
- 2

Scans transcribed for the reviewer

An image-only PDF is still escalated - but with the seven fields already transcribed by the model, marked as evidence, not a verdict.
- 3

Shipper history on the field

Correcting a field shows how often this shipper had a discrepancy on that same field before. A count, not a guess.
- 4

The original, one click away

Every value carries its line, and the source document opens beside it.
- 5

A reply drafted from the reviewer decision

Subject and body ready, built from what the reviewer decided - not the Sentinel result it overrode. A person still presses send.



The scan transcription, as the reviewer sees it: seven fields, the model's confidence, and the sentence that none of it entered the comparison.

Every flag carries the line it came from. Every escalation carries the reason and what to do about it. Most teams at this stage meet the brief; these are the things a reviewer actually uses.

Impact, and what comes next

11 hours → 13 seconds

520 emails, 124 document pairs. Eleven hours is a conservative estimate (20 s an email, 4 min a pair), not a measurement; thirteen seconds is measured. Every decision on that inbox was a rule's.

First deployment: one desk, not a region

The inbox already carries four desk codes (AFEMY 35, AIE 30, AFRT 29, AFPTME 22 emails). Sentinel sits beside the desk's existing check until the measures below have held for a full cycle of its carriers.

Measure	Today (graded inbox)	The pilot watches
Escalation rate	20 of 220 requests (9.1%), all correct	precision at 1.0 as unfamiliar templates grow
False discrepancies	0 of 46 discrepancies; 0 across 16 perturbation modes	the weekly number
Reviewer minutes per escalation	not measured yet	the pilot's first new measurement

Cheap because the model is aimed, not because it is absent.

Next, in order

- 1

Per-desk rules
a label table and an escalation policy per desk, selected by the code the inbox already carries
- 2

Corrections feed the label table
every confirmed correction is a labelled example of wording we could not read - the one place this system should learn
- 3

The database
one file; the demo does not need it yet, a pilot will

Every answer comes with its evidence.

Sentinel, by DuoCode

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